

Boris Knyazev

AI Research Scientist & Adjunct Professor

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RESEARCH AREAS

- Weight space learning
- Relational and graph-structured learning, graph neural networks (GNNs)
- Optimization, efficient training and inference of large (language) models (LLMs)
- Reasoning and scientific discovery

TEAM LEADING EXPERIENCE

- Mentoring and supervising of 10+ graduate students at Mila - Quebec AI Institute
- Leading internal Samsung projects on LLMs

EDUCATION

PhD, University of Guelph

Guelph (Ontario), Canada

Machine Learning Research Group

2017 - 2022

Thesis: Assessing and Improving Generalization in Graph Reasoning and Learning

Advisor: Graham W. Taylor

Teaching assistant: 2017-2019

Scholarships: IGTS, OGS, Dean's scholarship

MSc, Bauman Moscow State Technical University

Moscow, Russia

Diploma with Honors in Information Technology, *GPA: 4.8/5.0*

2005 - 2011

Scholarships: **German Academic Exchange Service (DAAD)**

PROFESSIONAL EXPERIENCE

University of Montreal

Montreal, Canada

Adjunct professor, DIRO (affiliation)

2025 - present

Samsung AI Lab

Montreal, Canada

Research Scientist (full time)

2022 - 2026

- Projects: LLMs, improving mixtures of experts, material discovery
- Manager: Simon Lacoste-Julien

Facebook AI Research

Montreal, Canada

Research Intern (full time)

Summer 2020

- Project: Parameter Prediction for Unseen Deep Architectures
- Advisors: Adriana Romero and Michal Drozdal

Mila - Quebec AI Institute

Montreal, Canada

Research Intern (full time)

2019 - 2020

- Project: Compositional Generalization in Scene Graphs
- Advisors: Eugene Belilovsky and Aaron Courville

NextAI

Toronto, Canada

Scientist in Residence (part time)

2018 - 2019

- Consulting startup teams

SRI International*Research Intern (full time)*

Princeton, NJ, USA

Summer & Fall 2018

- Project: Graph Neural Networks for Classification of Images and Other Modalities
- Advisor: Mohamed R. Amer
- Awards: Top-2 in all interns poster presentations

Institute for Neuro- and Bioinformatics, University of Luebeck

Luebeck, Germany

*Visiting researcher (full time)***2015 - 2016**

- Project: High-performance Image Classification with Small Training Datasets
- Advisors: Thomas Martinetz and Erhardt Barth

INVITED TALKS &
ORGANIZATIONS

- Co-organizing ICML 2026 Workshop on Weight-Space Symmetries: from Foundations to Practical Applications, <https://www.weightsymmetry.com/>.
- Invited talk “Neural Networks as Graphs” at ICLR 2025 Workshop on Weight Space Learning, <https://weight-space-learning.github.io/>.
- Interview with Yannic Kilcher about “Parameter Prediction for Unseen Deep Architectures” (YouTube, Nov 2021)

SELECTED
PUBLICATIONS

*equal contribution
complete list:
Google Scholar link

1. Li Y, Cloutier F, Wu S, Parviz A, **Knyazev B**, Zhang Y, Berseth G, Bang L. Mol4Gen: Multi-Agent, Multi-Stage Molecular Generation under Precise Multi-Property Constraints. **Under review**, 2025, arXiv.
2. Jha S, Hashemzadeh M, Pasand A.S., Parviz A, Lee M.J., **Knyazev B**. REAM: Merging Improves Pruning of Experts in LLMs. **Under review**, 2026, arXiv.
3. Moudgil A, **Knyazev B**, Belilovsky E. Celo2: Towards Learned Optimization Free Lunch. **ICLR**, 2026, arXiv.
4. Thérien B, Joseph C-E, **Knyazev B**, Oyallon E, Rish I, Belilovsky E. μ LO: Compute-Efficient Meta-Generalization of Learned Optimizers. **ICLR**, 2026 (also *oral at NeurIPS OPT Workshop 2024*), arXiv.
5. Jolicoeur-Martineau A, Baratin A, Kwon K, **Knyazev B**, Zhang Y. Any-Property-Conditional Molecule Generation with Self-Criticism using Spanning Trees. **TMLR**, 2025, arXiv.
6. Singh J, Misra D, **Knyazev B**, Orvieto A. (Almost) Free Modality Stitching of Foundation Models. **EMNLP** (Main track), 2025, arXiv.
7. **Knyazev B**, Moudgil A, Lajoie G, Belilovsky E, Lacoste-Julien S. Accelerating Training with Neuron Interaction and Nowcasting Networks. **ICLR**, 2025, arXiv.
8. Joseph C-E, Thérien B, Moudgil A, **Knyazev B**, Belilovsky E. Meta-learning Optimizers for Communication-Efficient Learning. **TMLR**, 2025, arXiv.
9. Moudgil A, **Knyazev B**, Lajoie G, Belilovsky E. Celo: Training Versatile Learned Optimizers on a Compute Diet. **TMLR (J2C Certification)**, 2025, arXiv.
10. Kofinas M, **Knyazev B**, Zhang Y, Chen Y, Burghouts G.J., Gavves E, Snoek C.G.M., Zhang D.W. Graph Neural Networks for Learning Equivariant Representations of Neural Networks. **ICLR (oral)**, 2024, arXiv.
11. **Knyazev B**, Hwang D, Lacoste-Julien S. Can We Scale Transformers to Predict Parameters of Diverse ImageNet Models? **ICML**, 2023, arXiv.
12. Schürholt K, **Knyazev B**, Giró-i-Nieto X, Borth D. Hyper-Representations as Generative Models: Sampling Unseen Neural Network Weights. **NeurIPS**, 2022, arXiv.
13. Thompson R, **Knyazev B**, Ghalebi E, Kim J, Taylor G.W. On Evaluation Metrics for Graph Generative Models. **ICLR**, 2022, arXiv.

14. **Knyazev B**, Drozdal M, Taylor G.W., Romero-Soriano A. Parameter Prediction for Unseen Deep Architectures. **NeurIPS**, 2021, arXiv.
15. Chung H, Kim J, **Knyazev B**, Taylor G.W., Lee J, Park J, Cho M. Brick-by-Brick: Sequential 3D Object Construction with Deep Reinforcement Learning. **NeurIPS**, 2021, arXiv.
16. **Knyazev B**, de Vries H, Cangea C, Taylor G.W., Courville A, Belilovsky E. Compositional Augmentations for Scene Graph Generation. **ICCV**, 2021, arXiv.
17. Lu Y, Rai H, Chang C, **Knyazev B**, Yu G, Shekhar S, Taylor G.W., Volkovs M. Context-aware Scene Graph Generation with Seq2Seq Transformers. **ICCV**, 2021, iccv.
18. **Knyazev B***, Augusta C*, Taylor G.W. Learning Temporal Attention in Dynamic Graphs with Bilinear Interactions. **PLOS One**, 2021, arXiv.
19. **Knyazev B**, de Vries H, Cangea C, Taylor G.W., Courville A, Belilovsky E. Graph Density-Aware Losses for Novel Compositions in Scene Graph Generation. **BMVC**, 2020, arXiv.
20. **Knyazev B**, Lin X, Amer M.R., Taylor G.W. Image Classification with Hierarchical Multi-graph Networks. **BMVC**, 2019, arXiv.
21. **Knyazev B**, Taylor G.W., Amer M.R. Understanding Attention and Generalization in Graph Neural Networks. **NeurIPS**, 2019 (also a *contributed talk*, 4% accept. rate, at *ICLR Workshop on Representation Learning on Graphs and Manifolds*), arXiv.